

Term Project Presentation

Energy-Robustness Trade-offs in Compressed TinyML Vision Models

CSE407 [Green Computing]

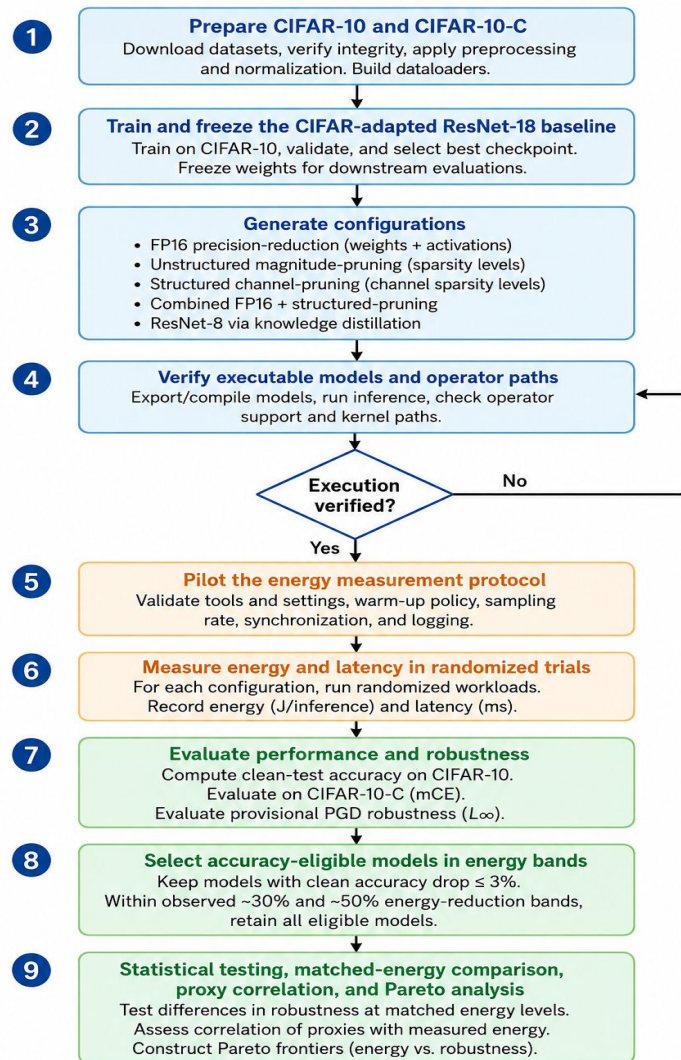
Section: 3 [Summer 2026]

Group: 07

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Methodology Overview

- Study compares five compression strategies using one CIFAR-10 ResNet-18 baseline.
- Evaluate three dimensions together: measured energy, clean accuracy, and CIFAR-10-C robustness.
- PGD is included only as a secondary/provisional robustness test.
- Primary comparison is based on measured energy reduction, especially around 30% and 50% targets.



Experimental Setup

- **Dataset: CIFAR-10: 50,000 training and 10,000 test images, 10 classes.**
- **CIFAR-10-C: 19 corruption types × 5 severity levels = 95 evaluation cells.**
- **Baseline: CIFAR-adapted ResNet-18, FP32.**
- **Hardware: NVIDIA Tesla T4; energy measured using NVML.**
- **Warm-up: ≥100 batches; 30 randomized trials targeting 5,000 images/model.**
- **NVML sampled every 4 ms; GPU synchronized before and after each measurement window.**

Dataset	CIFAR-10: 50,000 train / 10,000 test / 10 classes
Robustness set	CIFAR-10-C: 19 corruptions x 5 severities = 95 cells
Baseline	CIFAR-adapted ResNet-18, FP32
Hardware	NVIDIA Tesla T4
Energy backend	NVIDIA NVML; power sampled every 4 <u>ms</u>
Measurement	≥100-batch warm-up; 30 randomized trials; target 5,000 images
Primary metric	Measured joules per image

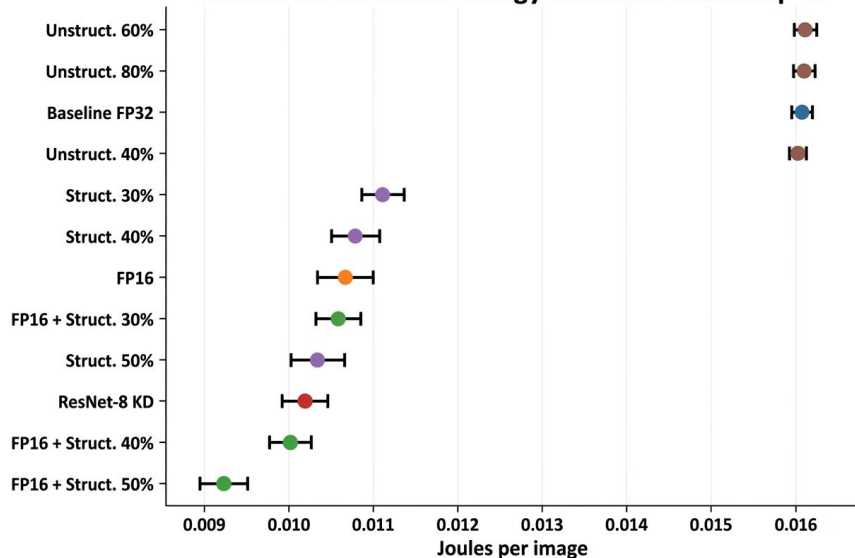
Compression Configurations

- **FP16 inference.**
- **Unstructured pruning: 40%, 60%, 80% sparsity; dense execution path retained.**
- **Structured channel pruning: 30%, 40%, 50%; dependent layers rebuilt.**
- **Combined: FP16 + structured pruning.**
- **Knowledge distillation: ResNet-8 student trained from the ResNet-18 teacher.**
- **Accuracy eligibility for matched-energy analysis: ≤ 3 percentage-point clean-accuracy loss.**

Strategy	Configurations	Execution / purpose
FP16	FP16 inference	Low-precision inference on T4
Unstructured pruning	40%, 60%, 80% sparsity	Dense execution path retained
Structured pruning	30%, 40%, 50% channel pruning	Channels removed; dependent layers rebuilt
Combined FP16 + structured	30% / 40% / 50%	Precision reduction + structural pruning
Knowledge distillation	ResNet-8 student	Smaller learned model distilled from ResNet-18

Main Result: Energy Reduction

Measured inference energy with 95% bootstrap CI

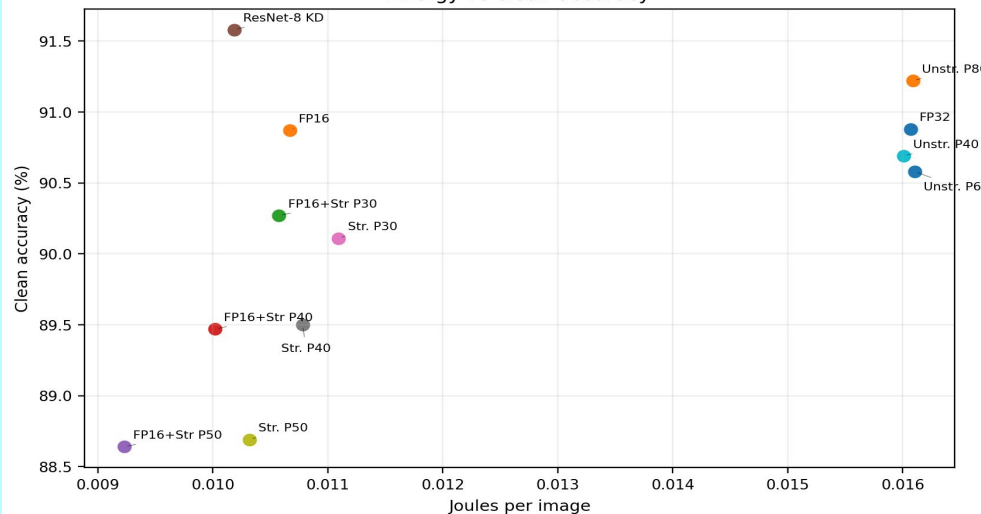


- Unstructured pruning gives essentially no measured energy benefit: +0.35% at 40% and slightly negative reductions at 60% and 80%.
- FP16, structured pruning, combined compression, and KD reduce energy by about 31–43%.

• Best measured reduction: FP16 + structured 50% = 42.56%.

• KD ResNet-8 reduces energy by 36.62%.

Energy vs clean accuracy



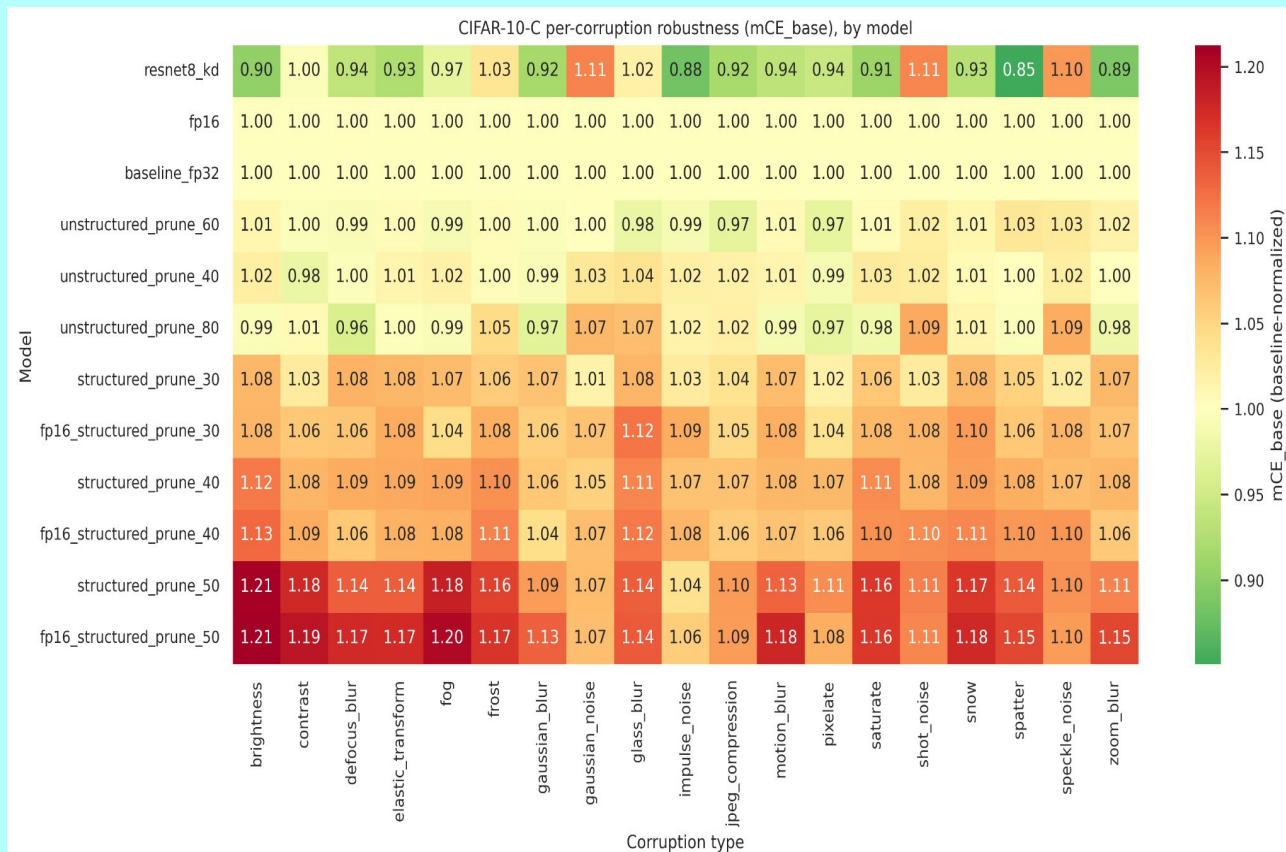
Energy vs. Accuracy: The Trade-off

- **Baseline accuracy: 90.88%.**
- **FP16: 90.87% accuracy with 33.60% lower energy, almost no accuracy cost.**
- **Structured 30%: 90.11% accuracy and 30.95% energy reduction.**
- **FP16 + structured 50%: 88.64% accuracy and 42.56% energy reduction.**
- **KD ResNet-8: 91.58% accuracy while using 36.62% less energy.**

Configuration	Energy reduction	Clean accuracy	Accuracy loss vs. baseline	mCEbase
FP16	33.60%	90.87%	0.01 pp	1.000
Structured 30%	30.95%	90.11%	0.77 pp	1.054
FP16 + Structured 50%	42.56%	88.64%	2.24 pp	1.144
ResNet-8 KD	36.62%	91.58%	+0.70 pp	0.962

CIFAR-10-C Robustness

- **FP16 preserves baseline corruption robustness: mCEbase = 1.000.**
- **Structured pruning worsens mCEbase as pruning increases: 1.054 → 1.083 → 1.130.**
- **Combined FP16 + structured pruning reaches 1.072–1.144.**
- **KD gives the best natural-corruption result: mCEbase = 0.962 with 91.58% clean accuracy.**
- **Main message: compression strategy changes robustness even when energy reduction is similar.**



Matched-Energy Comparison

Band	Model	Energy red.	Acc. loss	mCEbase	Interpretation
30% (25-35%)	FP16	33.60%	0.01 pp	1.000	Best robustness preservation
30% (25-35%)	Structured 30%	30.95%	0.77 pp	1.054	Higher corruption error
30% (25-35%)	FP16 + Structured 30%	34.19%	0.61 pp	1.072	Highest mCE in 30% band
50% (45-55%)	None	Not attained	-	-	Nearest eligible: 42.56%

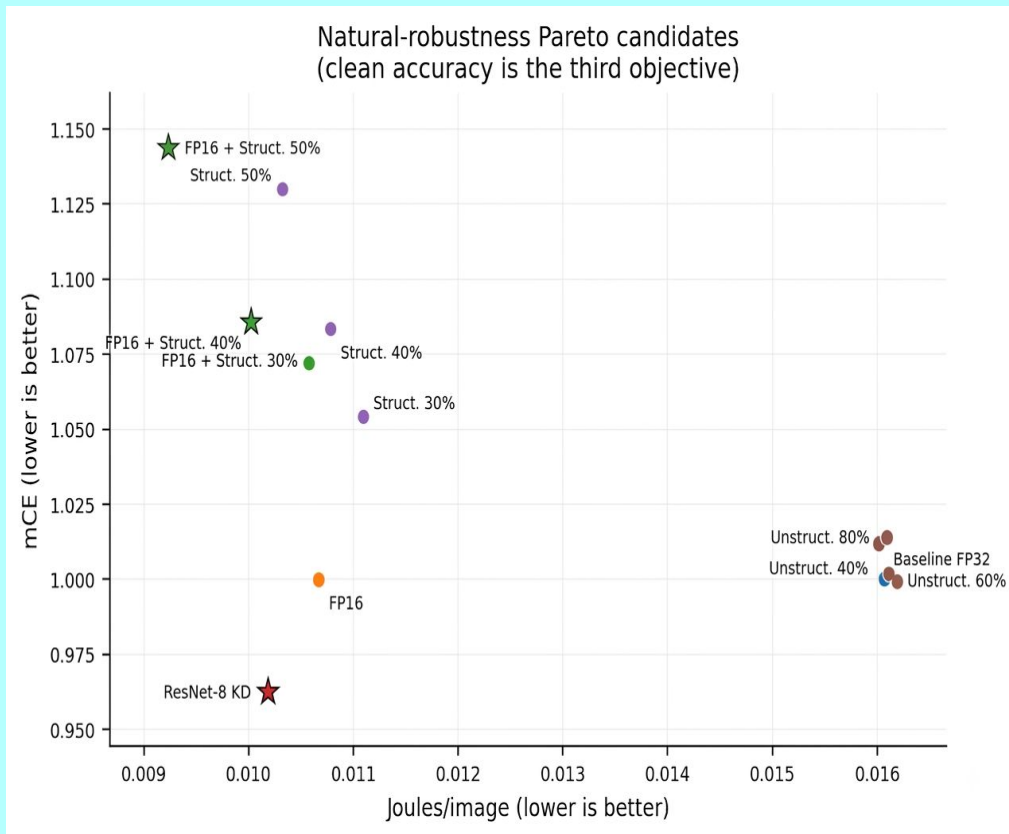
Target bands: 30% and 50% observed energy reduction, with ± 5 percentage-point tolerance.

- Only models within the 3-point clean-accuracy-loss limit are eligible.
- At the 30% target, three configurations qualify.

- FP16 gives the best robustness preservation in the 30% band.
- The 50% target was not attained within the accuracy eligibility limit.
- Therefore, the same energy budget can produce different robustness outcomes.

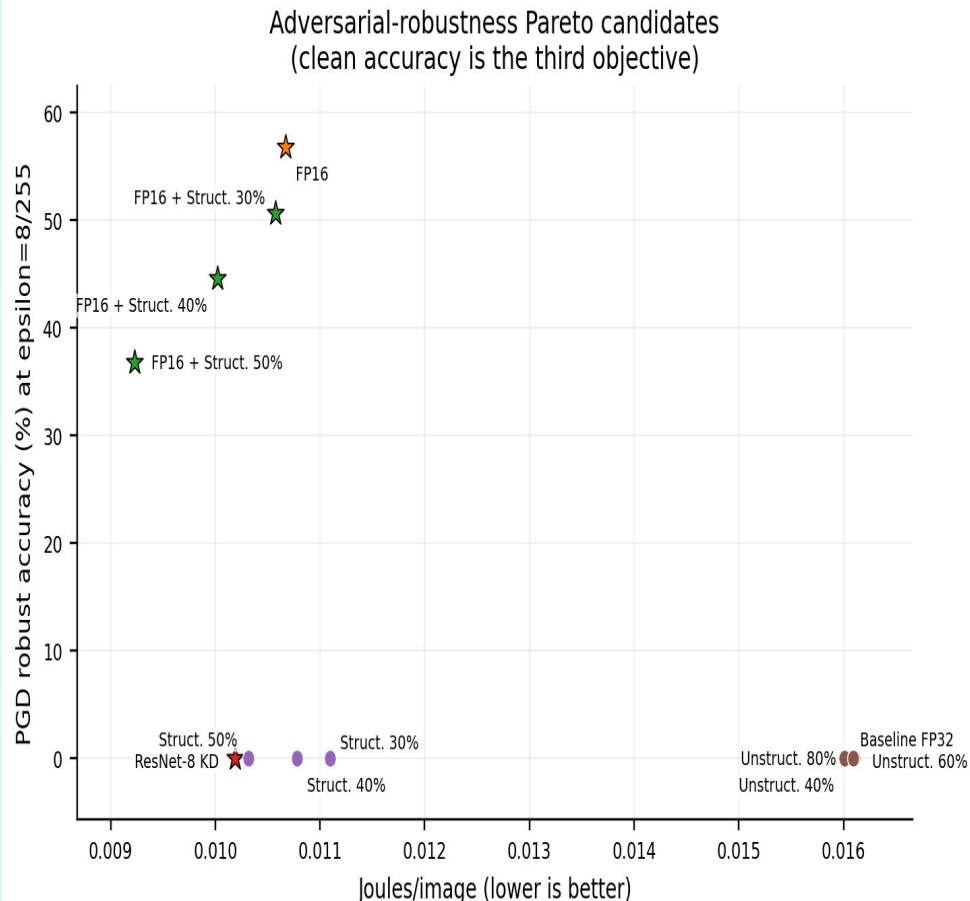
What Explains the Results?

- Measured energy depends more on executable structure and precision than on nominal sparsity.
- Unstructured pruning retains the dense execution path, so high nominal sparsity does not translate into energy savings on this GPU.
- Structured pruning changes the executable network and therefore produces real energy reduction.
- FP16 benefits from hardware-supported low-precision execution on the T4.
- KD behaves differently from post-hoc compression: the smaller learned architecture achieves a strong energy–accuracy–robustness trade-off.



Proxy Metrics vs. Measured Energy + PGD

- Measured energy strongly correlates with model size ($\rho=0.930$), latency ($\rho=0.951$), MACs ($\rho=0.835$), and parameter count ($\rho=0.828$).
- Nominal pruning ratio has weak correlation with measured energy ($\rho=0.274$, $\rho=0.476$).
- These are workload-specific correlations, not causal conclusions.
- PGD implementation failed its validity gate: FP32 configurations returned 0% robust accuracy while FP16 configurations returned non-trivial values.
- Therefore PGD is supportive/provisional only and is not used for the main adversarial ranking.



Key Contributions, Limitations & Future Work

- **Key contribution:** directly compares measured hardware energy with clean accuracy and natural-corruption robustness under matched experimental conditions.
- **Key finding:** nominal unstructured sparsity is a poor proxy for energy savings on the tested hardware.
- **Key finding:** FP16 is the strongest choice at the attainable ~30% energy-reduction level for preserving natural-corruption robustness.
- **Key finding:** KD ResNet-8 gives 36.62% lower energy, 91.58% clean accuracy, and best mCEbase = 0.962.
- **Limitations:** one dataset, one GPU/platform, one training seed, and provisional PGD results.
- **Future work:** validate PGD, evaluate more models/hardware platforms, and improve cross-model energy tracking.

References

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